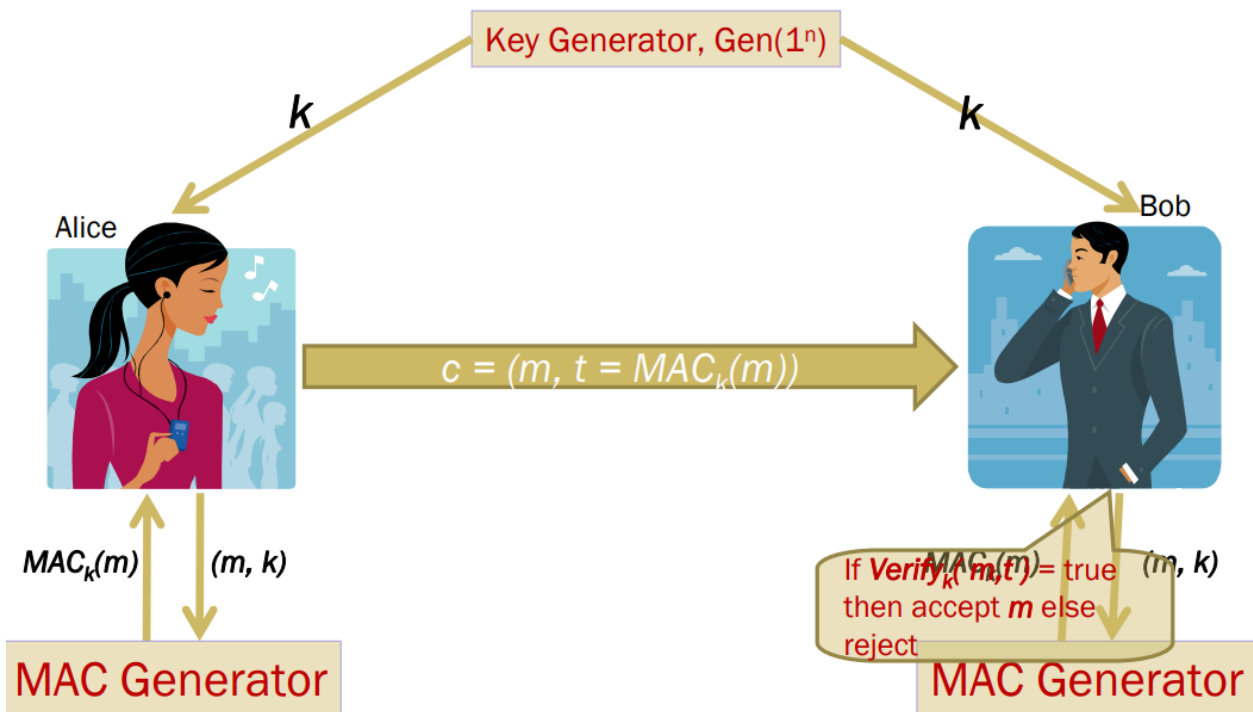


Message Authentication Code (MAC)

Theory:



- A Key Generation Algorithm that returns a secret key k
- A MAC generating algorithm that returns a tag for a given message m . Tag $t = MAC_k(m)$
- A Verification algorithm that returns a bit
- $b = Verify(m_1, t_1)$, given a message m_1 and a tag t_1
- If the message is not modified then with high probability, the value of b is true otherwise false

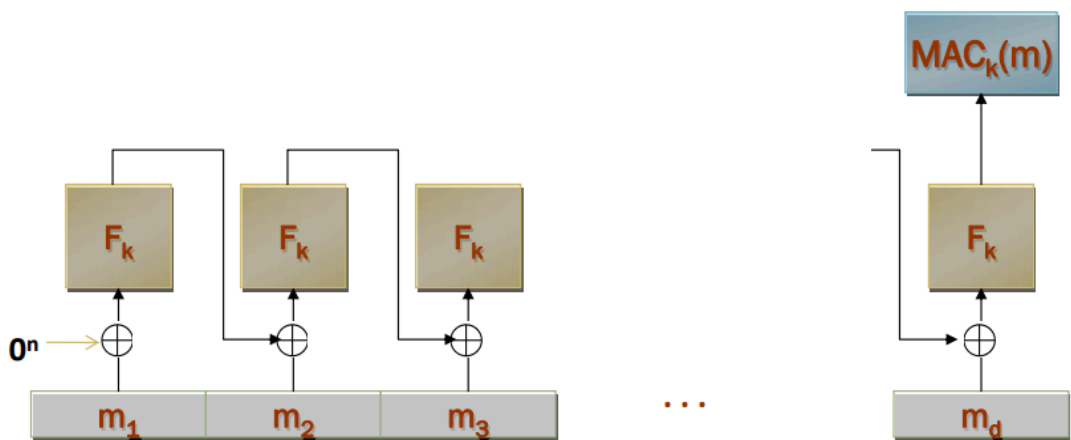
Generating MAC:

- Partition the message m to n sized blocks $m_1 m_2 \dots m_q$
- Calculate $MAC_k(m) = MAC_k(m_1 \oplus m_2 \dots \oplus m_q)$

Is this method secure?

NO! We are authenticating the xor of the message blocks but not the message itself. So we can always choose a message whose xor value is the same as some other message.

CBC-MAC:



Task: You are given 3 pieces of information: a message, key, and CBC-MAC signature. Your task is to verify whether the received message is valid or not.

Message	Key	MAC Signature	Validity
I met an interesting turtle while the song on the radio blasted away	b"\x01\xd8i\xa1^0\x9a<\x0f\x0r\xc1\xdd\xd5\x89\xa6'	ba4ecb8db45c6ae0	valid
I like to leave work after my eight-hour tea-break	b'\xa6+\x16\x9d-1\xda\x8aV\xed\x05\x0cv\x04\x88'	f47e78c537fa1435	
Her daily goal was to improve on yesterday	b'[\xc5\xbd\xe4z\xd1=E\x17-ku\x02= ='	ddaf3152edbe868a	
He found the chocolate covered roaches quite tasty	b'5"k\xff\x81a\x9b7\x8c>\xb7\xb9\xdcu\xaa'	9d30d856f84489a8	
After fighting off the alligator, Brian still had to face the anaconda	b'\xa1\xfcw"?3\x91\x1c\t\x9c\x91\xe2He\x935'	b9d173e05bbf7738	
He decided to count all the sand on the beach as a hobby	b'\xa7\x83@\xde\xbf\xb494\xee\x84\x1e-\xc8A\x09.'	6355e471bd9930a1	
The sign said there was road work ahead so he decided to speed up	b'2\xcbv\xdcU6\x99\xb6.\xa7\xea\xeb\xaf\x10\xc7\x90'	9fbafc75e0a5056a	
Send 500\$ to this account - 6589415651548	b'\xc3\xea\x99e\xaaI\xab\x04\x9b\x09\x0b4Z\x19\xed\xcf\xcb'	35273149636aca35	
Garlic ice-cream was her favorite	b'\x05\x09\x83\x9d\xb7\xb6\xc3\xb8\x9e\xc5\x09\x08\x07]\xc6\xb3'	dc2de1e07b71d391	
I'd rather be a bird than a fish	b'\x84YY\x00\x02GU\xa4LD\x05\x85!A\xc2c'	5e191d02aa5fc0b1	

Procedure:

Colab Notebook Link for this lab: [🔗 Lab 4 - CBC-MAC_A5_1](#)

- 1. Create a cmac object as shown using **key**
- 2. Update() the created object with your received message
- 3. Generate the MAC signature using finalize() function
- 4. Finally, print the decoded version of the signature and match it with your given signature.

A5/1

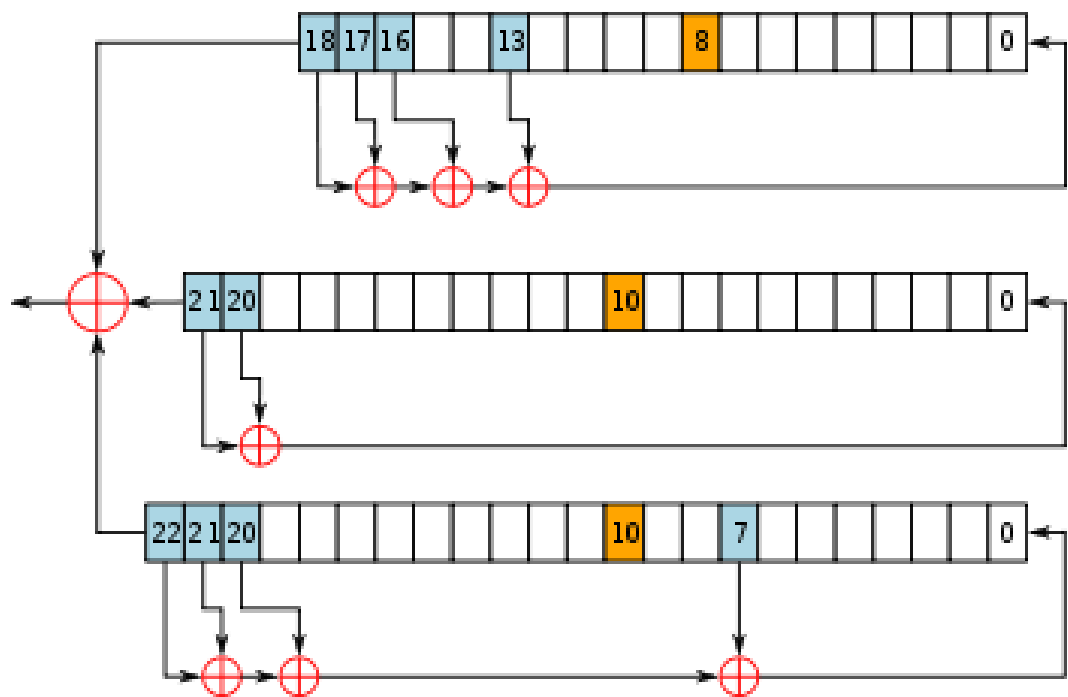
Theory:

A5/1 consists of 3 shift registers.

X: 19 bits

Y: 22 bits

Z: 23 bits



Procedure:

Encrypt the following plaintext:

X=1110001100101001011

Y=0011000000010000001101

Z=10011101101111001001110

Prepare a function **A51(X, Y, Z, n)**

Plaintext	Key stream len(Plaintext) Use A51 algorithm (Binary)	Ciphertext = Plaintext $\oplus$ key (Binary)	Plaintext = Ciphertext $\oplus$ Key (String)
It is alive			
Snap out of it			
I am as mad as hell and I am not going to take this anymore			
Bond James Bond			
Love means never having to say you're sorry			